

OBJECTIVES

- **Build Traffic Simulation** : Create a realistic urban intersection using SUMO with mixed vehicle types.
- **Monitor Traffic in Real Time** : Use TraCI and Python to detect vehicles, types, and waiting traffic.
- **Develop Smart Traffic Signals** : Dynamically control signal timing based on current traffic conditions (Type and Count).
- **Give Priority to Ambulances** : Detect approaching ambulances and provide them with a clear green–light path.
- **Evaluate System Performance** : Compare the adaptive system with fixed–time signals to measure improvements in traffic flow and waiting time.

